

MAJOR TEST SERIES - ENTHUSE + LEADER COURSE

PHASE : 1

TARGET : Pre-Medical 2025

Test Type : SEMI MAJOR

Test Pattern : NEET (UG)

TEST DATE : 24-11-2024

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	1	3	2	2	2	2	1	4	4	3	1	2	2	3	1	1	2	2	1	2	2	4	3	2	3	3	3	4	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	1	1	3	2	3	3	2	2	3	3	2	1	1	1	1	3	3	1	4	4	1	2	4	1	3	1	2	3	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	3	4	2	4	3	2	2	4	4	1	3	1	4	3	3	2	1	1	1	3	4	3	1	1	3	4	2	1	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	4	1	1	4	1	1	1	1	1	1	2	2	2	2	3	2	4	4	3	1	1	2	3	3	2	1	4	4	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	1	2	2	4	4	1	3	2	1	1	3	1	1	2	1	1	2	2	4	2	1	2	1	4	2	4	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	3	3	2	4	2	3	2	3	3	3	2	3	4	4	3	4	2	4	2	3	4	4	4	3	1	3	1	4	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	2	2	4	3	3	1	4	3	1	1	3	4	4	1	4	3	1	1	3										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. Ans (2)

Current within radius $r < R$:

$$I_{in} = I\pi r^2 = \frac{i}{\pi R^2} \pi r^2 = \frac{ir^2}{R^2}$$

$$\text{for } r < R, \oint \vec{B} \cdot d\vec{l} = \mu_0 I_{in} = \frac{\mu_0 ir^2}{R^2} \Rightarrow \text{Parabolic}$$

$$\text{for } r > R, \oint \vec{B} \cdot d\vec{l} = \mu_0 I_{in} = \mu_0 I \Rightarrow \text{Constant}$$

2. Ans (1)

$$\tau = I \alpha \quad \dots(1)$$

$$\tau = MB \quad \dots(2)$$

from (1) & (2)

$$(i\pi R^2)B = \left(\frac{mR^2}{2}\right)\alpha$$

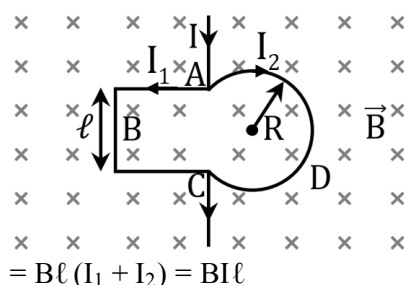
$$\alpha = \frac{2\pi iB}{m}$$

3. Ans (3)

The energy lost per unit volume of a substance in a complete cycle of magnetisation is equal to the area of the hysteresis loop.

4. Ans (2)

$$F = BI_1 \ell + BI_2 \ell$$



5. Ans (2)

$$Q = \frac{N\Delta\phi}{R} = \frac{\phi_2 - \phi_1}{R} = \frac{BA - 0}{R} = \frac{BA}{R}$$

6. Ans (2)

$$L = \frac{|e|}{di/dt} = \frac{12}{48/60} = 15H$$

7. Ans (2)

$$a = kt$$

$$\frac{dv}{dt} = kt$$

$$dv = kt dt$$

$$\int_u^v dv = k \int t dt$$

$$v - u = k \frac{t^2}{2}$$

$$v = u + \frac{kt^2}{2}$$

$$\frac{ds}{dt} = u + \frac{kt^2}{2}$$

$$\int ds = \int \left(u + \frac{kt^2}{2} \right) dt$$

$$s = ut + \frac{k}{2} \frac{t^3}{3}$$

$$s = ut + \frac{kt^3}{6}$$

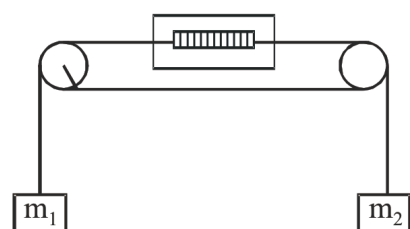
8. Ans (1)

$$F = u \left(\frac{dm}{dt} \right) = 3000 \times 4 = 12000 \text{ N}$$

9. Ans (4)

$$\text{Force} = m \frac{dv}{dt} = \frac{0.25 \times [(10) - (-10)]}{0.01} = 25 \times 20 = 500 \text{ N}$$

10. Ans (4)



$$\begin{aligned} \text{Reading} &= \text{Tension of spring} \\ &= \frac{2m_1 m_2 g}{m_1 + m_2} \end{aligned}$$

11. Ans (3)

$$\omega_h = \frac{2\pi}{12 \times 60 \times 60}$$

$$\omega_s = \frac{2\pi}{60}$$

$$\frac{\omega_h}{\omega_s} = \frac{1}{720}$$

12. Ans (1)

$T \propto R^2$, if radius becomes half then time period becomes $\frac{1}{4}$ of the previous value i.e. $\frac{24}{4} = 6$ hours.

Hence, the correct answer is option (a).

13. Ans (2)

$$\text{Diameter} = M. S. R. + (f \times L. C.)$$

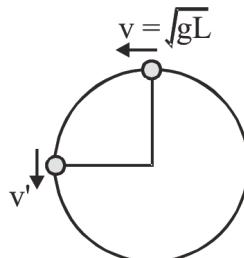
$$\text{Diameter} = 1 \text{ mm} + 42 \times \frac{1 \text{ mm}}{100}$$

$$\text{Diameter} = \frac{142}{100} \text{ mm}$$

$$\text{Diameter} = \frac{71}{50} \text{ mm}$$

$$\Rightarrow \text{So ; } x = 71$$

14. Ans (2)



$$v'^2 = v^2 + 2gL = gL + 2gL = 3gL$$

When the string is horizontal

$$a_c = \frac{v'^2}{L} = 3g$$

15. Ans (3)

$$W_{\text{ext}} = 0$$

$$W_{\text{int}} = \Delta KE = -\frac{1}{2} \frac{m_1 m}{(m_1 + m_2)} (\vec{v}_1 - \vec{v}_2)^2 \quad \text{OR}$$

$$K_f - K_i$$

$$W_{\text{int}} = \frac{1}{2} \times \frac{5 \times 5}{(5 + 5)} (2 - (-2))^2$$

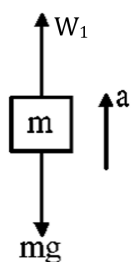
$$W_{\text{int}} = -\frac{1}{2} \times \frac{5}{2} \times 16$$

$$W_{\text{int}} = -20 \text{ J}$$

16. **Ans (1)**

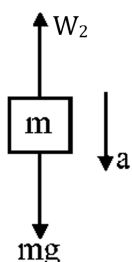
When lift moves up with constant acceleration a , then

$$w_1 - mg = ma \quad \dots(i)$$



When lift moves down with constant acceleration a , then

$$mg - w_2 = ma \quad \dots(ii)$$



From eqs. (i) and (ii), we get

$$w_1 + w_2 = 2mg \quad \dots(iii)$$

When lift moves up with constant speed, its acceleration is zero

$$\text{So, } w - mg = 0$$

$$w = mg \quad \dots(iv)$$

From eqs. (iii) and (iv), we get

$$w_1 + w_2 = 2w \Rightarrow w = \frac{w_1 + w_2}{2}$$

17. **Ans (1)**

For A

$$v_A = u - gt \quad (\text{in upward direction})$$

For B

$$v_B = gt \quad (\text{in downward direction})$$

As two balls are moving in opposite direction

$$v_{AB} = v_A + v_B = (u - gt) + gt = u$$

18. **Ans (2)**

$$e = \frac{v_2}{v_1} = \frac{\sqrt{h_2}}{\sqrt{h_1}} = \frac{2}{4} = \frac{1}{2}$$

19. **Ans (2)**

at $x = x_2$, $F = 0$

$$\frac{dF}{dx} < 0 \quad (\text{for stable equilibrium})$$

20. **Ans (1)**

By WET

$$W_g + W_{fr} = 0$$

$$mg \sin(30^\circ) \times 10 + W_{fr} = 0$$

$$W_{fr} = -mg \sin(30^\circ) \times 10$$

$$W_{fr} = -1 \times 10 \times \frac{1}{2} \times 10 = -50 \text{ J}$$

21. **Ans (2)**

$$v = -10t + 20$$

$$a = -10$$

$$F = -20$$

Displacement = Area under $v - t$ curve

$$= \frac{1}{2} \times 2 \times 20 = 20 \text{ m}$$

$$\text{Work done } \vec{F} \cdot \vec{ds} = -20 \times 20 = -400 \text{ J}$$

OR

WET

$$W = k_f^2 - k_i^2$$

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

here $v = 0$ $u = 20$ (By graph)

22. **Ans (2)**

For cylinder,

$$\frac{K^2}{R^2} = \frac{1}{2}$$

So, acceleration,

$$a = \frac{g \sin \theta}{1 + \frac{K^2}{R^2}} = \frac{g \sin 60^\circ}{1 + \frac{1}{2}} = \frac{g}{\sqrt{3}}$$

23. **Ans (4)**

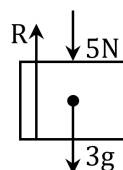
$$\alpha = \frac{\tau}{I} \Rightarrow \alpha = \frac{1000 \text{ N-m}}{200 \text{ kg-m}^2} \Rightarrow \alpha = 5 \text{ rad/sec}^2$$

Using 1st equation

$$\omega_f = \omega_i + \alpha t$$

$$\omega_f = 0 + (5)(3) = 15 \text{ rad/sec}$$

24. **Ans (3)**



F.B.D of block A

$$R = 5 + 3g$$

$$= 5 + 30$$

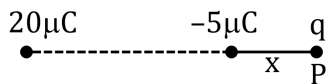
$$= 35 \text{ N}$$

25. **Ans (2)**



Let, charge q be placed at P.

At point P forces due to $20\mu\text{C}$ & $-5\mu\text{C}$ should be in opposite direction.



For net force $\vec{F} = 0$ & from coulomb's law force

$$F = \frac{kq_1q_2}{r^2} \Rightarrow k \frac{20q}{(5+x)^2} = \frac{k5q}{x^2} \Rightarrow x = 5\text{ cm}$$

26. **Ans (3)**

Inside the cavity net charge is zero.

$$\therefore Q_1 = 0 \text{ and } \sigma_1 = 0$$

There is no effect of point charges $+Q$, $-Q$ and induced charge on inner surface on the outer surface.

$$\therefore Q_2 = 0 \text{ and } \sigma_2 = 0$$

27. **Ans (3)**

For spherical shell

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (\text{if } r \geq R)$$

$$= 0 \quad (\text{if } r < R)$$

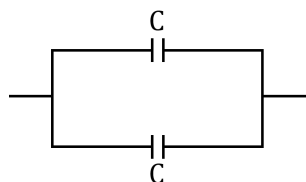
Force on charge in electric field, $F = qE$

$$\therefore F = 0 \quad (\text{For } r < R)$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} \quad (\text{For } r > R)$$

28. **Ans (3)**

The circuit can be reduced to



The equivalent capacitance of the combination is

$$C_{eq} = C + C = 2C$$

29. **Ans (4)**

Capacitance of sphere is given by :

$$C = 4\pi\epsilon_0 r$$

If, $C = 1\text{F}$ then radius of sphere needed :

$$r = \frac{C}{4\pi\epsilon_0} = \frac{1}{4\pi \times 8.85 \times 10^{-12}}$$

$$\Rightarrow r = \frac{10^{12}}{4\pi \times 8.85} = 9 \times 10^9 \text{ m}$$

$9 \times 10^9 \text{ m}$ is very large, it is not possible to obtain such a large sphere. Infact earth has radius $6.4 \times 10^6 \text{ m}$ only and capacitance of earth is $711 \mu\text{F}$.

30. **Ans (4)**

$$\text{Charge, } q = \int_{t_1}^{t_2} I \cdot dt = \int_0^{20} (20 + 3t) dt$$

$$= 20t + \frac{3t^2}{2} \Big|_0^{20} = 1000\text{C}$$

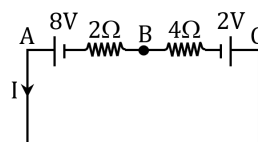
31. **Ans (4)**

In parallel combination, equivalent resistance,

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{\ell}{A} \cdot \frac{\rho_1 \rho_2}{\rho_1 + \rho_2} \left[\because R = \rho \frac{\ell}{A} \right]$$

$$\text{or, } R = \frac{25 \times 10^{-2}}{3 \times 10^{-6}} \times \frac{1.7 \times 2.6 \times 10^{-16}}{4.3 \times 10^{-8}} \therefore R = 0.858\text{m}\Omega$$

32. **Ans (1)**



$$I = \frac{8-2}{2+4} = \frac{6}{6} = 1\text{A}$$

Applying Kirchhoff voltage law, KVLC to B

$$V_C - 2 - 4 \times 1 = V_B$$

$$V_C - V_B = 6\text{V}$$

33. **Ans (1)**

They have high resistivity, low-temperature coefficient of resistance ability to attain good resistance values in small size.

34. **Ans (3)**

$$E = K + U = 100 + 200 = 300.0 \text{ J}$$

$$\Delta E = \Delta K + \Delta U = 2.0 + 1.0 = 3.0 \text{ J}$$

$$\% \frac{\Delta E}{E} = \frac{3.0}{300.0} \times 100 = 1\%$$

35. Ans (2)

L.C. = 1 MSD - 1 VSD(1)

& 1 MSD = 1 mm

χ MSD = N VSD

1 VSD = $\frac{x}{N}$ MSD

Now by equation (1)

$$\frac{1}{20N} \text{cm} = 1\text{mm} - \frac{x}{N}(1\text{mm})$$

$$\frac{10}{20N} \text{mm} = 1\text{mm} \left(1 - \frac{x}{N}\right)$$

$$\frac{x}{N} = 1 - \frac{1}{2N} = \frac{2N-1}{2N}$$

$$x = \left(\frac{2N-1}{2}\right)$$

SECTION-B

36. Ans (3)

Magnetic field at centre,

$$B = \frac{\mu_0 I}{2c} \otimes + \frac{\mu_0 I}{2b} \left(\frac{3}{4}\right) \odot + \frac{\mu_0 I}{2a} \left(\frac{3}{4}\right) \otimes$$

As per problem,

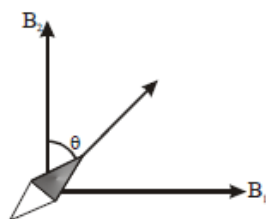
$$0 = \frac{\mu_0 I}{2c} \left[\frac{1}{c} - \frac{3}{4b} + \frac{3}{4a}\right] \Rightarrow \frac{3}{4} \left[\frac{1}{b} - \frac{1}{a}\right] = \frac{1}{c}$$

If $c = 2a$,

$$\frac{3}{4} \left[\frac{1}{b} - \frac{1}{a}\right] = \frac{1}{2a} \Rightarrow a = \frac{5}{3}b$$

37. Ans (3)

$$B_1 = \frac{\mu_0}{4\pi} \frac{2M}{d_1^3} \text{ (axial)}$$



$$B_2 = \frac{\mu_0}{4\pi} \frac{M}{d_2^3} \text{ (equatorial)}$$

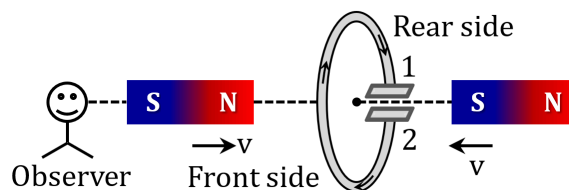
$$\frac{B_1}{B_2} = 2 \left(\frac{d_2}{d_1}\right)^3 = \tan \theta$$

$$\left(\frac{d_1}{d_2}\right)^3 = 2 \cot \theta$$

$$\frac{d_1}{d_2} = (2 \cot \theta)^{1/3}$$

38. Ans (2)

By the movement of both the magnets, current will be anticlockwise, as seen from left side, i.e., plate 1 will be positive and 2 will be negative.



39. Ans (2)

$P^x Q^y c^z$ is dimensionless

$$[ML^{-1}T^{-2}]^x [MT^{-3}]^y [LT^{-1}]^z = [M^0L^0T^0]$$

$$[M^{x+y} L^{-x+z} T^{-2x-3y-z}] = [M^0L^0T^0]$$

Comparing powers of M, L and T

$$x + y = 0 \Rightarrow y = -x$$

$$-x + z = 0 \Rightarrow z = x$$

$$-2x - 3y - z = 0$$

x and y are equal and of opposite sign

x and z are equal and of same sign

$$x = 1, y = -1, z = 1$$

40. Ans (3)



centripetal acceleration

41. Ans (3)

$$\text{Loss in P.E.} = mgh = 1.8 \times 10^5 \times 10 \times 50$$

$$= 900 \times 10^5 \text{ J}$$

Energy used to generate power

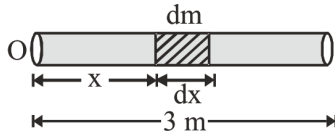
$$= \frac{900 \times 10^5}{2 \times 3600} = 12500 \text{ W}$$

$$\text{Number of bulbs} = \frac{12500}{100} = 125$$

42. Ans (2)

Mass per unit length, i.e. linear mass density

$$\mu \propto x \Rightarrow \mu = \lambda x, \lambda : \text{constant}$$



$$dm = \mu dx = \lambda x dx$$

$$x_{c.m.} = \frac{\int x dm}{\int dm} = \frac{\int_0^3 x \lambda x dx}{\int_0^3 \lambda x dx}$$

$$= \frac{\int_0^3 x^2 dx}{\int_0^3 x dx} = \frac{\left[\frac{x^3}{3}\right]_0^3}{\left[\frac{x^2}{2}\right]_0^3} = \frac{\frac{3^3}{3}}{\frac{3^2}{2}} = 2 \text{ m}$$

43. Ans (1)

According to parallel axes theorem,

$$I_{AB} = \frac{2}{5} M \left(\frac{R}{2}\right)^2 + M(2R)^2 + \frac{2}{5} M \left(\frac{R}{2}\right)^2$$

$$I_{AB} = 4MR^2 + \frac{1}{5} M(R)^2 = \frac{21}{5} MR^2$$

Hence, the correct answer is option (1).

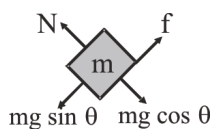
44. Ans (1)

Distance increase continuously.

45. Ans (1)

$$N = mg \cos \theta, f = mg \sin \theta$$

Net force applied by M on m (or m on M) :

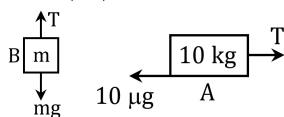


$$F = \sqrt{N^2 + f^2}$$

$$= \sqrt{(mg \cos \theta)^2 + (mg \sin \theta)^2}$$

$$= mg$$

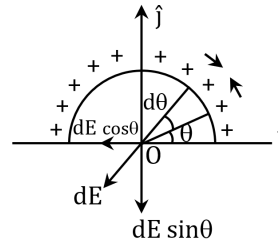
46. Ans (1)



$$T = mg \Rightarrow T \geq \mu 10g$$

$$\Rightarrow mg \geq 0.20 \times 10g \Rightarrow m \geq 2kg$$

47. Ans (3)



Let us consider a differential element dl subtending at angle $d\theta$ at the centre O as shown in the figure. Linear charge density

$$\lambda = \frac{q}{Qr}$$

Charge on the element

$$dq = \left(\frac{q}{\pi r}\right) dl$$

$$= \frac{q}{\pi r} (rd\theta) = \left(\frac{q}{\pi}\right) d\theta (\because dl = rd\theta)$$

Electric field at the centre O due to dq is

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\pi r^2} d\theta$$

Resolving dE into two rectangular component, we find the component $dE \cos \theta$ will be counter balanced by another element on left portion. Hence resultant field at O is the resultant of the component $dE \sin \theta$ only.

$$\therefore E = \int dE \sin \theta = \int_0^\pi \frac{q}{4\pi^2 r^2 \epsilon_0} \sin \theta d\theta$$

$$= \frac{q}{4\pi^2 r^2 \epsilon_0} [-\cos \theta]_0^\pi = \frac{q}{4\pi^2 r^2 \epsilon_0} (+1 + 1) = \frac{q}{2\pi^2 r^2 \epsilon_0}$$

The direction of E is towards negative y -axis

$$\therefore \vec{E} = -\frac{q}{2\pi^2 r^2 \epsilon_0} \hat{j}$$

48. Ans (3)

When switch is closed then both capacitor will be parallel.

$$C_{eq} = 2C$$

$$\text{Energy } E_1 = \frac{1}{2} C_{eq} V^2 = \frac{1}{2} \times 2C \times V^2$$

$$E_1 = CV^2$$

When switch is opened charge on right capacitor remain CV while potential on left capacitor remain same.

Dielectric $K = 5$

$$C' = KC$$

$$C' = 5C$$

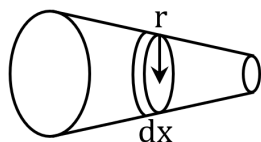
$$E_2 = \frac{1}{2} C' V^2 + \frac{q^2}{2C'}$$

$$\Rightarrow E_2 = \frac{1}{2} (5C) V^2 + \frac{(CV)^2}{2(5C)}$$

$$E_2 = \frac{5CV^2}{2} + \frac{CV^2}{10} \Rightarrow E_2 = \frac{13CV^2}{5}$$

$$\frac{E_1}{E_2} = \frac{CV^2}{\frac{13CV^2}{5}} = \frac{5}{13} \Rightarrow \frac{E_1}{E_2} = \frac{5}{13}$$

49. Ans (1)



Current is constant in the conductor, $I = \text{constant}$

$$\text{Resistance of element of conductor, } dR = \frac{\rho dx}{\pi r^2}$$

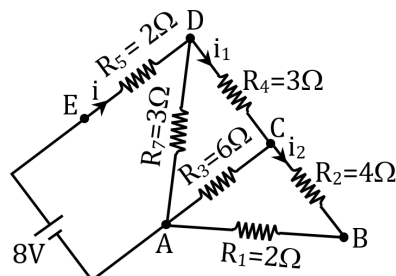
$$dV = i dR = \frac{i \rho dx}{\pi r^2}$$

$$E = \frac{dV}{dx} = \frac{i \rho}{\pi r^2}$$

Drift velocity,

$$V_d = \frac{eE\tau}{m} \quad \text{As } V_d \propto E \text{ and } E \propto \frac{1}{r^2}$$

50. Ans (4)



Equivalent resistance,

$$R_{eq} = 4\Omega$$

$$\text{Current, } i = \frac{8}{4} = 2A$$

$$\text{Current through } R_4, i_1 = \frac{2 \times 3}{3 + 6} = \frac{2}{3}A$$

$$\text{Current through } R_2, i_2 = \frac{2/3}{2} = \frac{1}{3}A$$

SUBJECT : CHEMISTRY

SECTION-A

51. Ans (4)

Hydrazine ($\text{NH}_2\text{—NH}_2$) have no carbon so does not show Lassaigne's test.

52. Ans (1)

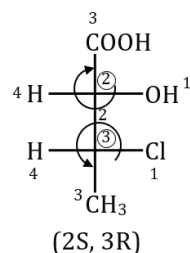
The most common polar and acidic support used is adsorption chromatography is silicagel. The surface silanol groups adsorb polar compound and work particularly well for basic substances. Alumina is the example of polar and basic adsorbent that is used in adsorption chromatography.

53. Ans (2)

There are two stereo centers ($n = 2$)

$$\text{No. of stereoisomer} = 2^n = 2^2 = 4$$

54. Ans (4)

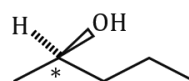


Clockwise Rotation = "S"

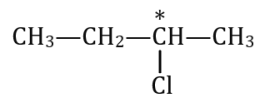
Anticlockwise Rotation = "R"

(2S, 3R)

56. Ans (3)



Chiral centre



Chiral centre

Optically active = 2

57. Ans (1)

In Vinylic halide halogen atom bonds with carbon atom of $\text{C}=\text{C}$ bond.

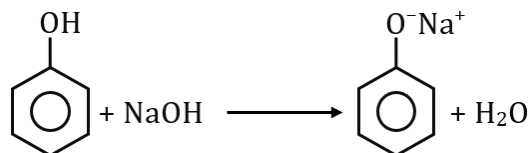
58. Ans (2)

sp^2 hybridized carbocation $\left[\text{CH}_3\text{—CH}=\overset{\oplus}{\text{C}}\text{H} \right]$ is very unstable, therefore $\text{CH}_3\text{—CH}=\text{CH—Cl}$ can not give $\text{S}_{\text{N}}1$ reaction.

59. Ans (3)

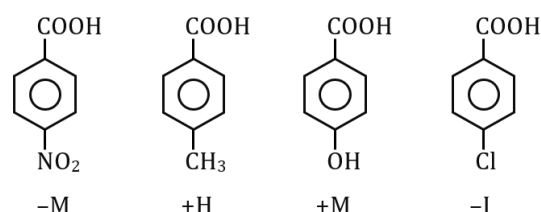
In $\text{S}_{\text{N}}1$ reaction carbocation is formed. The carbocation formed at bridge head is very unstable due to restricted bond angle. So, it is inactive towards $\text{S}_{\text{N}}1$

60. Ans (4)



Stronger acid than H_2O due to more resonating structure of phenoxide ion.

61. Ans (1)



62. **Ans (3)**

Addition of HCN to carbonyl compound is nucleophilic addition reaction.

63. **Ans (4)**

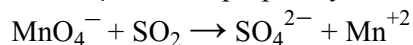
Aldehyde (aliphatic and aromatic) show reaction with Tollen's reagent and aliphatic aldehydes show Fehling solution, but Ketones do not show there reactions.

64. **Ans (2)**

Aminobenzene and aniline are same organic compounds.

65. **Ans (4)**

KMnO₄ has dark purple crystals.



MnO₄⁻ in Mn⁺⁷ has d⁰ configuration, still it shows colour because of ligand to metal charge transfer.

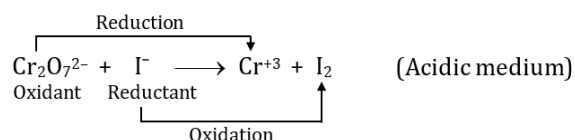
66. **Ans (3)**

Y, Ta belongs to d-block, Yb is a part of lanthanide series.

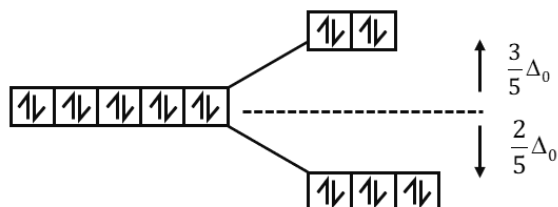
67. **Ans (2)**

Tungsten is having the highest melting point in the entire 5d series.

68. **Ans (2)**



69. **Ans (4)**



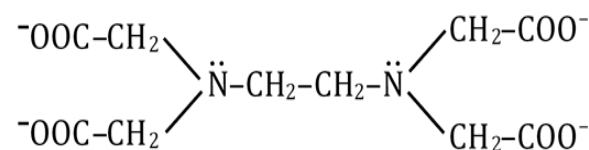
$$\begin{aligned} \text{CFSE} &= 6 \times \frac{-2}{5} \Delta_0 + \frac{3}{5} \Delta_0 \times 4 \quad \text{or} \quad (-0.4 \times 6 + 0.6 \times 4) \Delta_0 = \text{CFSE} \\ &= \frac{-12}{5} \Delta_0 + \frac{12}{5} \Delta_0 = 0 \quad \Rightarrow \quad (-2.4 + 2.4) \Delta_0 = 0 \end{aligned}$$

70. **Ans (4)**

[Ma₃b₃] type shows fac and mer.

71. **Ans (1)**

EDTA (ion)



72. **Ans (3)**

SiC is a covalent carbide.

73. **Ans (1)**

As we move down in a periodic table, the lower oxidation state becomes more stable due to inert pair effect

In 13th group, order of relative stability of +1 oxidation state is Al < Ga < In < Tl

74. **Ans (4)**

NF₃, CCl₄ → No V.O.

SF₆ → Steric Hinderance

75. **Ans (3)**



Here $\frac{n}{m} = \frac{5}{2} = 2.5$ S - S bond will present here.

77. **Ans (2)**

$$t_{75\%} = 2 \times t_{1/2}$$

80. **Ans (1)**

Higher the SRP, Better the oxidising agent.

81. **Ans (3)**

For sucrose; $\pi_1 = \text{CRT}$... (i)

For AB₃; $\pi_2 = i \times \text{CRT}$... (ii)

Divide (i) and (ii) we get

$$\frac{0.24}{0.72} = \frac{1}{i} \\ i = 3$$

$$i = 1 + (n-1)\alpha$$

$$3 = 1 + (4-1)\alpha$$

$$\alpha = 2/3$$

82. Ans (4)

SPM allows only solvent molecules to pass through it.

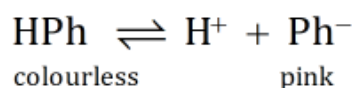
83. Ans (3)

$$\frac{P_o - P_s}{P_o} = \frac{in_2}{n_1 + in_2} \quad \frac{10}{50} = \frac{3 \times n}{3n + 12}$$

84. Ans (1)

Protective power of lyophilic colloid

85. Ans (1)



SECTION-B

86. Ans (3)

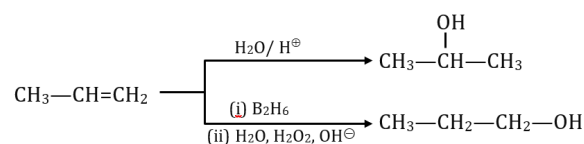
(I) Lucas reagent-Conc. HCl/ZnCl₂

(II) Dumas method-CuO/CO₂

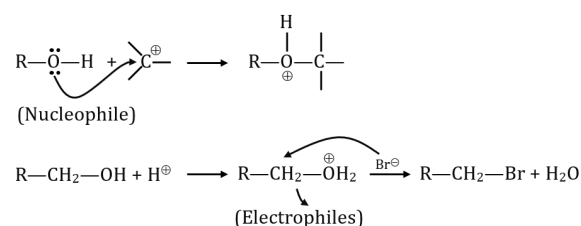
(III) Kjeldahl's method-H₂SO₄

(IV) Hinsberg test-C₆H₅SO₂Cl/aq.KOH

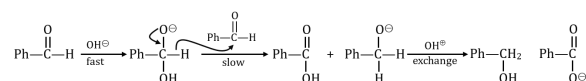
87. Ans (4)



88. Ans (2)



89. Ans (1)

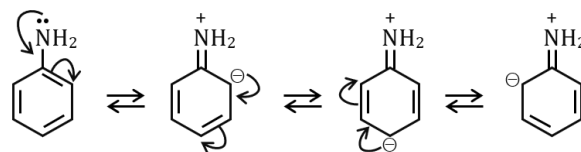


90. Ans (3)

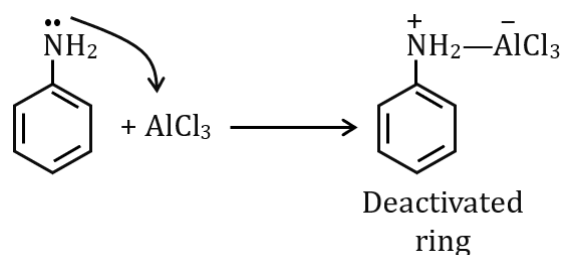
Only those aldehyde which do not have α-H atom undergo Cannizzaro's reaction. Hence CH₃CHO will not undergo Cannizzaro reaction.

91. Ans (1)

-NH₂ is an ortho and para directing group, in activates benzene ring towards electrophilic substitution reaction.



Aniline does not undergo Friedel-Crafts reaction due to salt formation with AlCl₃ which deactivate ring towards electrophilic substitution reaction.



92. Ans (4)

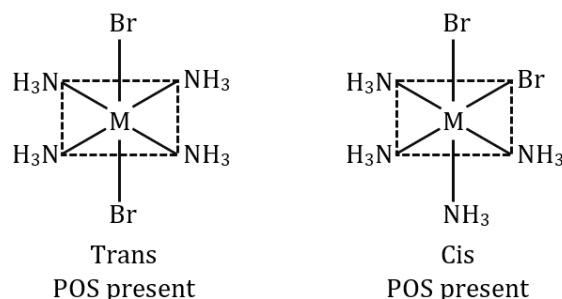
KMnO₄ can act as oxidising agent in all the mediums – neutral, acidic and alkaline media.

MnO₄⁻ → Mn⁺² (acidic medium)

MnO₄⁻ → MnO₂ (neutral medium)

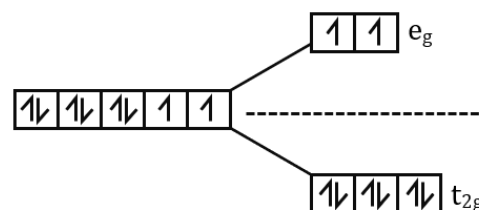
MnO₄⁻ → MnO₄²⁻ (alkaline medium)

93. Ans (1)



94. Ans (1)

Ni²⁺(3d⁸4s⁰), NH₃(SFL)



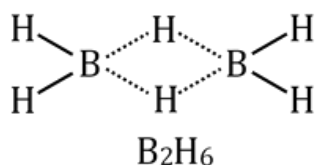
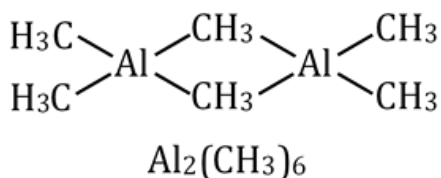
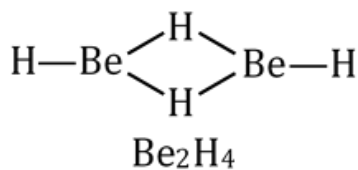
Unpaired e⁻ = 2

$$\text{So, Magnetic moment} = \sqrt{n(n+2)}$$

$$= \sqrt{2(2+2)} = \sqrt{8} = 2.83 \text{ BM}$$

95. Ans (4)

$3C - 2e^-$ bond is present in respectively.



96. Ans (1)

$$R = K_1[\text{H}_2][\text{I}]^2 [\text{slow}], \quad \dots(i)$$

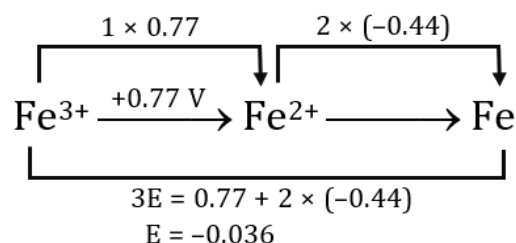
$$K_{\text{eq}} = \frac{[\text{I}]^2}{[\text{I}_2]} \Rightarrow [\text{I}]^2 = K_{\text{eq}} [\text{I}_2] \quad \dots(ii)$$

putting ii into i we get option (1)

97. Ans (1)

Higher the SRP, Lesser the reducing power.

98. Ans (1)

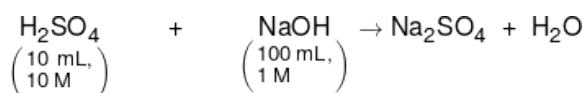


Which means it exist is oxidised form.

99. Ans (1)

i increases, So higher the elevation in Boiling point.

100. Ans (1)



$$\begin{array}{l} n_{\text{gram eq.}} \quad 10 \times 2 \times 10 \quad \quad 100 \times 1 \\ \quad \quad \quad = 200 \text{ mg eq.} \quad \quad 100 \text{ mg eq.} \end{array}$$

$$M_{\text{eq}} \text{ of Acid} > M_{\text{eq}} \text{ of Base}$$

SUBJECT : BIOLOGY-I

SECTION-A

104. Ans (2)

NCERT-XI, Pg. # 61

105. Ans (2)

NCERT-XI, Pg. # 62

106. Ans (3)

NCERT-XI, Pg. # 61,62,66

107. Ans (2)

NCERT-XI, Pg. # 61

108. Ans (4)

NCERT-XI, Pg. # 76

109. Ans (4)

NCERT-XI, Pg. # 73

111. Ans (1)

NCERT-XI, Pg. # 105

112. Ans (1)

NCERT-XI, Pg. # 105

115. Ans (3)

NCERT Page # 139

116. Ans (2)

NCERT Page # 138

117. Ans (1)

NCERT Page # 140

121. Ans (1)

NCERT-XI, Pg. # 13

125. Ans (2)

NCERT-XI, Pg. # 27

127. Ans (4)

NCERT-XI, Pg. # 32

129. Ans (3)

NCERT-XI, Pg. # 94

131. Ans (1)

NCERT-XI, Pg. # 123

132. Ans (1)

NCERT-XI, Pg. # 126

SECTION-B

141. **Ans (4)**
NCERT Page # 139
145. **Ans (1)**
NCERT-XI, Pg. # 22
147. **Ans (2)**
NCERT-XI, Pg. # 88
148. **Ans (4)**
NCERT-XI, Pg. # 95
149. **Ans (3)**
NCERT-XI, Pg. # 126

SUBJECT : BIOLOGY-II

SECTION-A

151. **Ans (1)**
NCERT, Pg. # 44, Para-4.2.8
152. **Ans (3)**
NCERT, Pg. # 38, Para-4.1.1
153. **Ans (3)**
NCERT Page No.- 102(old) ;
154. **Ans (2)**
NCERT Page No.- 101(old);
155. **Ans (4)**
NCERT Page No. 109
156. **Ans (2)**
NCERT-XI, Pg # 189
157. **Ans (3)**
NCERT-XI, PG. #186
158. **Ans (2)**
NCERT-XI, PG. #190
159. **Ans (3)**
NCERT XI Pg. # 195
160. **Ans (3)**
NCERT Pg # 196
161. **Ans (3)**
NCERT, Pg. # 196, Para-15.1.3.2

162. **Ans (2)**
NCERT XI, Pg.# 297-298
163. **Ans (3)**
NCERT PG 291
164. **Ans (4)**
New NCERT-XI, Pg. # 218
165. **Ans (4)**
NCERT, Pg # 225
166. **Ans (3)**
NCERT Pg#236
167. **Ans (4)**
NCERT Pg. # 319, 321
168. **Ans (2)**
NCERT Page # 321
169. **Ans (4)**
NCERT-XI, Pg # 230,231
170. **Ans (2)**
Posterior pituitary gland is not a true endocrine gland because it only stores and releases hormones.
171. **Ans (3)**
NCERT Page # 342
172. **Ans (4)**
NCERT Page # 333,336
173. **Ans (4)**
NCERT, Pg. # 339
174. **Ans (4)**
NCERT Page # 57
175. **Ans (3)**
NCERT Page # 48
176. **Ans (1)**
NCERT Page # 47
177. **Ans (3)**
NCERT Page # 111
178. **Ans (1)**
NCERT Page # 276

179. **Ans (4)**
NCERT Page # 284
180. **Ans (4)**
ATP is utilized to contract muscle (Mechanical process)
181. **Ans (2)**
NCERT Page # 321
182. **Ans (2)**
NCERT Page # 339
183. **Ans (2)**
NCERT Page # 337
184. **Ans (4)**
NCERT Page # 294
185. **Ans (3)**
NCERT Page # 236

SECTION-B

186. **Ans (3)**
NCERT Pg.#44-46.
187. **Ans (1)**
NCERT, Pg. # 50, 4.2.11.7
188. **Ans (4)**
NCERT, Pg. # 47, Para-4.2.11
189. **Ans (3)**
NCERT Pg. # 79

190. **Ans (1)**
NCERT-XII, Pg. # 187
191. **Ans (1)**
NCERT XI Pg.# 282 - 283
192. **Ans (3)**
NCERT Pg # 209, 210
193. **Ans (4)**
NCERT Page no 297
194. **Ans (4)**
NCERT Page No.- 105(old);
195. **Ans (1)**
NCERT XI Pg.# 236
196. **Ans (4)**
NCERT-XI Pg. # 245, 246
197. **Ans (3)**
NCERT-XI, Pg. # 243
198. **Ans (1)**
NCERT Page # 111
199. **Ans (1)**
NCERT Page # 270
200. **Ans (3)**
NCERT Page # 287,288